Author Contributions Checklist Form

This form documents the artifacts associated with the article (i.e., the data and code supporting the computational findings) and describes how to reproduce the findings.

Part 1: Data

☐ This paper **does not** involve analysis of external data (i.e., no data are used or the only data are generated by the authors via simulation in their code).

☒ I certify that the author(s) of the manuscript have legitimate access to and permission to use the data used in this manuscript.

Abstract

The breast cancer data used in the paper are publicly available from The Cancer Genome Atlas Program (TCGA, https://tcga-data.nci.nih.gov/tcga/). The data were downloaded using R package "TCGA-Assembler" (https://github.com/compgenome365/TCGA-Assembler-2).

Availability

☒ Data **are** publicly available

☐ Data **cannot be made** publicly available

If the data are publicly available, see the *Publicly available data* section. Otherwise, see the *Non-publicly available data* section, below.

Publicly available data

☒ Data are available online at: https://tcga-data.nci.nih.gov/tcga/

☒ Data are available as part of the paper's supplementary material.

☐ Data are publicly available by request, following the process described here:

☐ Data are or will be made available through some other mechanism, described here:

Non-publicly available data

Discussion of lack of publicly available data:

Description

File format(s)

☐ CSV or other plain text:
☒ Software-specific binary format (.Rda, Python pickle, etc.): .RData files
☐ Standardized binary format (e.g., netCDF, HDF5, etc.):
☐ Other (described here):

Data dictionary

☒ Provided by the authors in the following file(s): “Readme.pdf” under folder “PxG_code”
☐ Data file(s) is (are) self-describiing (e.g., netCDF files)
☐ Available at the following URL:

Additional information (optional)

The mRNA data contain 883 samples corresponding to three genes: ERBB2, ESR1 and PGR. The protein data contain 883 samples corresponding to 15 genes: PTEN, PIK3CA, BAD, NFKB1, TSC1, IRS1, EIF4EBP1, EGFR, AKT1, TSC2, MTOR, DVL3, ERBB2, ESR1, PGR.

Part 2: Code

Abstract

All codes are available under the subfolder “Code” that includes five R scripts, one C++ script, one zip file (another model for comparison) and one R package (another model for comparison). “PxG-GW.R” and “PxG-PL.cpp” contain all functions for the PxG model with Gaussian-G-Wishart and pseudo-likelihood implementation, respectively. “Example1.R” produces all results and figures in section 4.1; “Example2.R” produces all results and figures in section 4.2; “Example3.R” produces all results and figures in section 4.3; “RealDataAnalysis.R” produces all results and figures in section 5. “LSGraph_0.1.tar.gz” is the R package for the BM13 model from Bhadra & Mallick (2013) and Niu et al. (2020); “JRNS.zip” contains three C++ scripts for the JRNS model from Samanta et al. (2022); mSSL package can be downloaded from GitHub, see the corresponding R code in “Example2.R” and “Example3.R”. All three models are needed for model comparison with the PxG model. “RealDataAnalysis.R” requires breast cancer data from subfolder “Data”.

Description

Code format(s)

☒ Script files
- ☒ R ☐ Python ☐ Matlab
- ☒ Other: C++

☐ Package
- ☐ R ☐ Python ☐ MATLAB toolbox
- ☐ Other:

☒ Reproducible report
- ☒ R Markdown ☐ Jupyter notebook
- ☐ Other:

☐ Shell script

☐ Other (described here):

Supporting software requirements

Version of primary software used

The analysis and simulation studies used R version 4.2.1 (2022-06-23).
Platform: arm64-apple-darwin22.4.0 (64-bit)
Hardware: Apple M1 Max with 10 cores and 32 GB memory
Running under: macOS Ventura 13.3.1

Libraries and dependencies used by the code

R libraries required:
huge, invgamma, BDgraph, ggplot2, ggpubr, devtools, mSSL (from GitHub), SSLASSO, MASS, LSGraph (included in the "Code" subfolder), Rcpp, RcppArmadillo, gglasso, glasso, glmnet, mvtnorm, rms, dlm.

Supporting system/hardware requirements (optional)

Parallelization used

☒ No parallel code used
☐ Multi-core parallelization on a single machine/node
Number of cores used:
☐ Multi-machine/multi-node parallelization
Number of nodes and cores used:

License

☒ MIT License (default)
☐ BSD
☐ GPL v3.0
☐ Creative Commons
☐ Other (described here):

Additional information (optional)

Part 3: Reproducibility workflow

Scope

The provided workflow reproduces:

☒ Any numbers provided in text in the paper

☒ The computational method(s) presented in the paper (i.e., code is provided that implements the method(s))

☒ All tables and figures in the paper

☐ Selected tables and figures in the paper, as explained and justified here:

Workflow details

Format(s)

☐ Single master code file

☐ Wrapper (shell) script(s)

☒ Self-contained R Markdown file, Jupyter notebook, or other literate programming approach

☒ Text file (e.g., a readme-style file) that documents workflow

☐ Makefile

☐ Other (more detail in 'Instructions' below)

Instructions

A. Before running any R scripts, install all required R packages listed above. To reproduce all simulation studies and real data analysis using R scripts in the subfolder “Code”:

A.1. To reproduce all simulation results and figures in section 4.1, directly run “Example1.R” in the subfolder “Code”.

A.2. To reproduce all simulation results and figures in section 4.2, first install mSSL package (see line 152-154 in “Example2.R” for instructions) and install LSGraph package (see line 195-196 in “Example2.R” for instructions), then run “Example2.R” in the subfolder “Code”.

A.3. To reproduce all simulation results and figures in section 4.3, first install mSSL package (see line 221-223 in “Example3.R” for instructions) and install LSGraph package (see line 268-269 in “Example3.R” for instructions), then unzip “JRNS.zip” for three C++ scripts ("JRNS.cpp",

"StepB.cpp", "StepOmega.cpp") required for the JRNS model. Finally, run “Example3.R” in the subfolder “Code”.

A.4. To reproduce all results and figures in section 5, first load breast cancer data (“BRCA_RNA.RData”, “BRCA_RPPA.RData”) from the subfolder “Data”, then run “RealDataAnalysis.R” in the subfolder “Code”.

B. Before running any R Markdown files, install all required R packages listed above. To reproduce simulation studies and real data analysis under **ONLY** the PxG model using R Markdown files in the subfolder “RMarkdown”, first copy “PxG_GW.R” and “PxG_PL.cpp” from the subfolder “Code” to the subfolder “RMarkdown”, copy “BRCA_RNA.RData” and “BRCA_RPPA.RData” from the subfolder “Data” to the subfolder “RMarkdown”, then:

B.1. To reproduce simulation results and figures under only the PxG model in section 4.1, directly run “Example1.Rmd” in the subfolder “RMarkdown”.

B.2. To reproduce simulation results and figures under only the PxG model in section 4.2, directly run “Example2.Rmd” in the subfolder “RMarkdown”.

B.3. To reproduce simulation results and figures under only the PxG model in section 4.3, directly run “Example3.Rmd” in the subfolder “RMarkdown”.

B.4. To reproduce results and figures under only the PxG model in section 5, directly run “RealDataAnalysis.Rmd” in the subfolder “RMarkdown”.

For more details, see “Readme.pdf” in the folder “PxG_code”.

Expected run-time

Approximate time needed to reproduce the analyses on a standard desktop machine:

☐ <1 minute

☐ 1-10 minutes

☐ 10-60 minutes

☒ 1-8 hours

☐ >8 hours

☐ Not feasible to run on a desktop machine, as described here:

Additional documentation (optional)

Notes (optional)